



BHABHA UNIVERSITY, BHOPAL

Programme : DIPLOMA in Mechanical Engineering(ME)

Session : 2018-19

Scheme of Studies and Examinations for : FIRST SEMESTER

COURSE CODE	COURSE TITLE	THEORY COMPONENT					PRACTICAL COMPONENT			GRAND TOTAL OF MARKS
		LECTUR ES					PRACTI CAL Hrs. Per Week	CONTINUOUS EVALU ATION		
			Hrs. Per Week	TERM WORK QUIZ, ASSIGN MENT	MID TERM TEST (TWO)				LAB. WORK QUIZ, ASSIGN MENT	
I	II	MARKS			MARKS					
HU-1001	COMMUNICATION SKILLS	06	10	10	10	70	02	20	30	150
PY-1001	PHYSICS	06	10	10	10	70	02	20	30	150
CH-1001	CHEMISTRY	06	10	10	10	70	02	20	30	150
MA-1001	MATHEMATICS	06	10	10	10	70	--	--	--	100
FU-1001	FUNDAMENTAL OF ELECTRICAL & CIVIL ENGINEERING PRACTICE	-	-	-	-	-	02	40	60	100
PA-1001	PROFESSIONAL ACTIVITIES (PA) & HUMAN VALUE (HV)	--	--	--	--	--				
	TOTAL	24	40	40	40	280	8	100	150	650

L: Lecture

T: Tutorial

P: Practical

End semester examination will be of 3 hour duration

**SYLLABUS
FOR DIPLOMA IN:
MECHANICAL ENGINEERING (ME)**

(FIRST SEMESTER)



**Scheme: 2018
Implemented from session 2018-19**



BHABHA UNIVERSITY, BHOPAL

DIPLOMA IN MECHANICAL ENGINEERING

SEMESTER: FIRST

SYLLABUS

COURSE CODE: HU-1001

NAME OF COURSE:

COMMUNICATION SKILLS

RATIONALE

The wide range of communicative and functional need of English in the evolving global and technical professional environment has more than ever imposed a demand of acquiring proficiency in communication skills in our technicians and diploma pass outs. Besides being a professional language, it also acts as a window to technical and scientific knowledge. Diploma pass outs are required to communicate with personnel belonging to different echelons of authority. Therefore, acquiring proficiency in listening, speaking, reading and writing English is an integral part of professional and technical competence.

Upon completion of this course, the student will be able to: Understand slowly delivered spoken material in Indian English. Understand general purpose words of English.

Use general purpose words of English to express himself in speaking reasonably clearly and correctly on routine matters.

Write reasonably and grammatically correct English.

Develop a habit of reading with comprehension to achieve an optimum speed of 75 WPM. Communicate effectively in a professional environment through speaking and writing to achieve desired objectives.



BHABHA UNIVERSITY, BHOPAL

DIPLOMA IN MECHANICAL ENGINEERING

SEMESTER: **FIRST**
COURSE CODE: HU-1001
NAME OF COURSE: **COMMUNICATION SKILLS**

SYLLABUS

Lectures: 6 Hrs. per week Practical: 2 Hrs. per week

SYLLABUS OF STUDY

S. No.	Topic
1	COMMUNICATION PROCESS AND ITS NEEDS
2	PASSAGES OF COMPREHENSION
3	BUSINESS COMMUNICATION (ONE QUESTION WITH INTERNAL CHOICE)
4	COMPOSITION & TRANSLATION
5	UNSEEN PASSAGES & PRECIS WRITING



BHABHA UNIVERSITY, BHOPAL

DIPLOMA IN MECHANICAL ENGINEERING

SEMESTER: **FIRST**

SYLLABUS

COURSE CODE: HU-1001

NAME OF COURSE: **COMMUNICATION**

SKILLS

Lectures: 6 Hrs. per week Practical: 2 Hrs. per week

COURSE CONTENTS

S.NO.	TOPIC	CONTENTS
1	COMMUNICATION	1.1 (i) How to make communication effective
	PROCESS AND ITS NEEDS	(ii) Barriers in communication, Removal of barriers (i) Determiners, Prepositions, Auxiliary verbs and subject-verb (iii) Vocabulary – One word
02.	PASSAGES OF COMPREHENSION	substitution, words often misused and 2.1 Prescribed passages (six from existing syllabus) i Language of Science ii Desalination or Desalting Process iii Safety Practices iv Non-conventional Sources of 2.2 Writing summary, moral and characterization of any one story from the
3	BUSINESS COMMUNICATION (One question with internal choice)	3.1 Principles of effective business correspondence Its parts, mechanics, styles and forms

S.NO.	TOPIC	CONTENTS
04.	COMPOSITION & TRANSLATION	3.2 Application for job, Bio-Data and C.V. 3.3 Letter of Enquiry Placing order Complaint 4.1 Writing paragraphs of 150 words on topics of general interest i.e. pollution, ragging in college, importance of computers, importance of communication skill, importance of 4.2 science and technology etc. Translation (Hindi to English and vice- versa).
05.	UNSEEN PASSAGES & PRECIS WRITING	i Answer the questions based on the passage. ii Give suitable title OR Writing Precis



BHABHA UNIVERSITY, BHOPAL

MECHANICAL ENGINEERING

SEMESTER: FIRST

SYLLABUS

COURSE CODE: HU-1001

NAME OF COURSE: COMMUNICATION SKILLS

REFERENCES

1. English Conversation Practice
- Grant Taylor
2. Practical English Grammar
- Thomson & Martinet
3. Communication Skills for Technical Students Book – I, Book – II
- M/S Somaiya Publication, Bombay
4. Living English Structure
- S. Allen
5. English Grammar, Usage, and Composition
- Tickoo & Subramanian, S. Chand & Co. Standard Allen Longman.
6. Essentials of Business Communication
- Dr. Rajendra Pal & J.S. Korlahalli S.Chand & Sons, New Delhi.
7. Effective Business Communication
- M.V. Rodriques, Concept Pub. Co., New Delhi.
8. Communication for Business
- Shirely Taylor, Longman, England.
9. Communication for Engineers and Professors
- P. Prasad, S.K.Kataria and sons publications, New Delhi
10. Technical English Book-II,
- Somaya Publications, New Delhi



BHABHA UNIVERSITY, BHOPAL
DIPLOMA IN
MECHANICAL ENGINEERING

SEMESTER: **FIRST**

SCHEME: **2018**

COURSE CODE: **HU-1001**

NAME OF COURSE: **COMMUNICATION SKILLS**

SUGGESTED READINGS

S.No.	SUGGESTED READINGS
1	<u>To enhance the reading skills and generate interest</u> ➤ A Brief History of Time: - Stephan Hawking, Bentham Books, Great Britain ➤ Cosmos: - Carl Sagan, Bentham Books, Great Britain. ➤ Ignited Minds: - A.P.J. Abdul Kalam, Penguin Books. ➤ India 2020: - A.P.J. Abdul Kalam, and Y.S. Rajan Penguin Books. ➤ Beyond the Last Blue Mountain: - J.R.D. Tata, Penguin Books ➤ Life and Times: - Albert Einstein, Bentham Books. ➤ Power of Oration: - Abraham Lincoln.
2	<u>Faster reading for deriving Pleasure.</u> ➤ Interpreter of Maladies: - Jhumpa Lahiri., Harper & Collins. ➤ Short stories by R.K.Narayan, Tagore, Tolstoy, Mulkraj Anand, O. Henry.
3	<u>For Vocabulary Building.</u> ➤ Word Power made Easy: - Norman Lewis, Bloomsbury ➤ Reading, Spelling, Vocabulary, Pronunciation, Book 1,2 &3: - Norman Lewis. ➤ The Joy of Vocabulary: - Levine, Levine & Levine. ➤ Roget's Thesaurus of Synonyms and Antonyms. ➤ Cambridge English Pronouncing Dictionary: - Daniel Jones . ➤ Audio- Visual learning resources and multimedia learning material for pronunciation improvement and listening skills.



BHABHA UNIVERSITY, BHOPAL

MECHANICAL ENGINEERING

SEMESTER: FIRST
COURSE CODE: PY-1001
NAME OF COURSE: PHYSICS

SCHEME: 2012
PAPER CODE: 6031

RATIONALE

Curriculum revision needs to be updated and revised in the light of the changes occurring in the life so that they fulfill the objectives.

1. To minimize the poor technical knowledge in the basics of his / her discipline.
2. To improve practical skill on the basis of theoretical knowledge imported.
3. To improve the problem solving skill.

Physical science forms the foundation of engineering, the subject of physics has its importance amongst all the physical sciences, therefore, it is to be taught exclusively to the students of diploma in engineering.



BHABHA UNIVERSITY, BHOPAL

DIPLOMA IN MECHANICAL ENGINEERING

SEMESTER: **FIRST**

COURSE CODE: PY-1001

NAME OF COURSE: **PHYSICS**

Lectures: **6** Hrs. per week

Practical: **2** Hrs. per week

SCHEME OF STUDIES

S. No.	Topic
1	UNITS & MEASUREMENT
2	MOTION
3	MOLECULAR PHENOMENON OF SOLIDS, LIQUIDS AND GASES
4	PROPERTIES OF MATTER
5	HEAT
6	HEATING EFFECT OF CURRENT AND THERMOELECTRICITY
7	SOUND
8	OPTICS AND OPTICAL INSTRUMENTS
9	ELECTROSTATICS AND ELECTROMAGNETIC INDUCTION
10	MODERN PHYSICS, BASIC ELECTRONICS



BHABHA UNIVERSITY, BHOPAL

DIPLOMA IN MECHANICAL ENGINEERING

SEMESTER: **FIRST**
COURSE CODE: **PY-1001**
NAME OF COURSE: **PHYSICS**

SCHEME: 2018

Lectures: **6 Hrs.** per week

COURSE CONTENTS

S.NO.	TOPIC	CONTENTS
1	UNITS & MEASUREMENT	Fundamental and derived units Scalar and vector, Basic requirements to represent vector Symbols, abbreviation, and proculation Linear measurement by vernier calipers, screw gauge and spherometer
2.	MOTION	Angular measurement by angular vernier Motion and its type Linear motion (laws and equation) Circular motion Angular velocity and relation with linear velocity Centripetal acceleration, Centripetal and Centrifugal forces Rotatory motion Axis of rotation Moment of Inertia, Radius of gyration Kinetic energy of rotation Numerical problems and solution on the topic
3.	MOLECULAR PHENOMENON OF SOLIDS, LIQUIDS AND GASES	Postulates Of Molecular Kinetic Theory Of Structure of matter Brownian motion Kinetic and Potential energy of molecules Kinetic theory of gases Postulates

S.NO.	TOPIC	CONTENTS
4.	PROPERTIES OF MATTER	Calculation of pressure by Kinetic theory Prove of different gases law by Kinetic theory. Elasticity: Meaning, definition, stress, strain, Hook's law and elastic limit Surface Tension : Meaning, definition, molecular forces, cohesive and adhesive forces, surface energy, capillary rise and capillary rise method. Viscosity : Meaning, definition, stream line and turbulent flow, critical velocity, Stock's law. Numerical problems and solution on the topic
5.	HEAT	Heat and temperature, concept of heat as molecular motion Transmission of heat, steady state and variable state. Concept of heat capacity, specific heat and latent heat. Calorimeter and its uses Thermodynamics
6.	HEATING EFFECT OF CURRENT AND THERMOELECTRICITY	Relation between heat and work Mechanical equivalent of heat First law of thermodynamics and its application Second law of thermodynamics and its application Carnot cycle Numerical problems and solution on the topic. Heating effect of electric current: Joule's law, work energy and power in electric circuit, calculation of electric energy. Thermo electricity Seeback effect and thermoelectric power. Neutral temperature, temperature of inversion and relation between them Thermo electric thermometer and thermo couples.

S.NO.	TOPIC	CONTENTS
7.	SOUND	Numerical problems and solution on the topic. Production of sound waves(Longitudinal and transverse waves) Progressive and stationary waves
8.	OPTICS AND OPTICAL INSTRUMENTS	Basic knowledge of refraction , reflection, interference and diffraction.Ultrasonic,Audible range, Production of ultrasonic, properties and uses Refraction, critical angle and total internal reflection, refraction through lenses and problems Power of lenses, Spherical and chromatic aberrations.
9.	ELECTROSTATICS AND ELECTROMAGNETIC INDUCTION	telescope and derivation for their magnifying power, Numerical problems and solution on the topic.Coulomb's law, Electric field intensity, potential. Capacity, principle of capacitor, types of capacitor, combination of capacitors Electromagnetic Induction: Faraday's law, Lenz's law Self and mutual inductance Transformer and electric motor, Induction coil
10.	MODERN PHYSICS,	Photoelectric effect, threshold P-N junction, diode as rectifier.



BHABHA UNIVERSITY, BHOPAL

DIPLOMA IN MECHANICAL ENGINEERING

SEMESTER: **FIRST**
COURSE CODE: PY-**1001**
NAME OF COURSE: **PHYSICS**

SCHEME: 2018

Practical: 2 Hrs. per week

LIST OF EXPERIMENTS

S.NO.	NAME OF THE EXPERIMENT
1	Refractive index of prism (I-d) curve
2	Refractive index of prism (spectrometer)
3	Focal length of a convex lens by u-v method
4	Focal length of a convex lens by displacement method
5	Verification of Ohm's law
6	To find out unknown resistance by meter bridge
7	To find out internal radius of hollow tube by vernier calipers. To find out volume of given cylinder by screw gauge.
8	Surface tension by Capillary rise method. Coefficient of viscosity
9	Coefficient of Thermal conductivity by searl's method.
10	Verification of Newton's cooling law.



BHABHA UNIVERSITY, BHOPAL

**DIPLOMA IN
MECHANICAL ENGINEERING**

SEMESTER: **FIRST**
COURSE CODE: PY-1001
NAME OF COURSE: **PHYSICS**

SCHEME: **2018**

REFERENCES

1. APPLIED PHYSICS VOL. 1 & 2
- SAXENA AND PRABHAKAR
2. PHYSICS
- TTTI PUBLICATION
3. PHYSICS VOL. 1 & 2
- HALLIDAY AND RESNIC R
4. ENGINEERING PHYSICS
- GAUR AND GUPTA
5. PRINCIPLE OF PHYSICS
- BRIJ LAL & SUBRAMANYAN
6. PHYSICS FOR TECHNICAL EDUCATION
- LS ZEDNOV



BHABHA UNIVERSITY, BHOPAL
DIPLOMA IN
MECHANICAL ENGINEERING

SEMESTER: **FIRST**
COURSE CODE: **1001**
NAME OF COURSE: **CHEMISTRY**

SCHEME: **2018**

Lectures: **6** Hrs. per week

Practical: **2** Hrs. per week

SCHEME OF STUDIES

S. No.	Topic
1	ATOMIC STRUCTURE AND RADIOACTIVITY
2	SURFACE CHEMISTRY AND ITS APPLICATION
3	ELECTROCHEMISTRY
4	COLLIGATIVE PROPERTIES
5	CHEMICAL BONDING AND CATALYSIS
6	WATER
7	METALS AND ALLOYS
8	IONIZATION, PH VALUE CORROSION AND PROTECTION
9	GLASS, CEMENT AND REFRACTORY
10	HIGH POLYMERS, RUBBER AND INSULATORS
11	LUBRICANTS, PAINTS AND VARNISHES
12	FUELS, FIRE EXTINGUISHERS AND EXPLOSIVES
13	POLLUTION AND CONTROL



BHABHA UNIVERSITY, BHOPAL
DIPLOMA IN
MECHANICAL ENGINEERING

SEMESTER: **FIRST**

COURSE CODE: CH-1001

NAME OF COURSE: **CHEMISTRY**

SCHEME: **2018**

Lectures: **6 Hrs.** per week

COURSE CONTENTS

S.NO.	TOPIC	CONTENTS
1	ATOMIC STRUCTURE AND RADIOACTIVITY	Discovery of electron, proton ,neutron and nucleus. Ruther ford's and Bohr's model of an atom. Bohr Burry scheme of filling the electrons in various orbits. Idea of s,p,d,f orbital .Alfa, Gamma and Beta rays, theory of radio activity, Group displacement law, half life period, numerical problems on half life period, fission and fusion.
2.	SURFACE CHEMISTRY AND ITS APPLICATION	True solution, colloidal solution and suspension, lyophobic and lyophillic colloids, optical and electrical properties of colloids, coagulation, coagulants, idea about gels and emulsions.
3.	ELECTROCHEMISTRY	Electrolysis, Faraday's laws of electrolysis, Numerical problems on Faradays Law, electroplating of copper and nickel.
4.	COLLIGATIVE PRPOPERTIES	Osmosis & osmatic pressure, Relative vapour pressure and Routls law. Internal energy (enthalpy) Entrophy, Entrophy fusion free energy, Effect of change in temperature catalysis.
5.	CHEMICAL BONDING AND CATALYSIS	(A) Bonding: Nature of bonds- Electrovalent, Co-valent, co-ordinate and hydrogen bond.

S.NO.	TOPIC	CONTENTS
6.	WATER	(B) Catalysis : Types , theory characteristic, positive , negative, auto and induced catalyst. Catalytic Promoter, and catalytic inhibitors. Industrial Application of catalysis. Sources of water, types of water, hardness of water, its causes, types and removal, Boiler feed water, harmful – effects of hard water in boiler. Municipal water supply. Numerical on soda lime process. Determination of hardness of water by O. Hener's, EDTA and soap solution method
7.	METALS AND ALLOYS	Physical and chemical properties of metals, copper, iron, aluminum, tin, nickel. General principal of metallurgy, minerals/ ores, ore dressing, roasting ,smelting, bassemerisation, fluxes, purification . Explanation of alloying purposes, methods of alloying, composition and uses of alloy like brass, bronze, duralium, German silver, gun metal, solder, stainless steel, casting and bearing alloy.
8.	IONIZATION, PH VALUE CORROSION AND PROTECTION	Arhenius theory of ionization, factors affecting ionization. pH meaning (numerical), Buffer solutions and Buffer actions, choice of indication (acidimetry and alkalimetry). Explanation of corrosion, types of corrosion, factors effecting corrosion, corrosion control (protection against corrosion), metal and organic coating for corrosion control.
9	GLASS, CEMENT AND REFRACTORY	Glass: Basic raw materials for glass, composition and manufacture of glass, varieties of glass and annealing of glass,. Cement : Constituting compounds in cement, Composition of Portland Cement, its manufacture, setting and hardening of cement.Refractories : Meaning, characteristics , use of common refractory materials.
10.	HIGH POLYMERS, RUBBER AND INSULATORS	Polymerization and condensation, classification of plastics, Compounding and Moulding constituents of plastics. Preparation Properties and uses of PVC, polyethene, polystyrene, polyamides, polyesters , Bakelite. Synthetic fibers –

S.NO.	TOPIC	CONTENTS
11.	LUBRICANTS, PAINTS AND VARNISHES	nylon, rayon, decron, and polyesters. Definition characteristics , classification and properties of insulators. Glass, wool and thermocole. Idea about rubber and vulcanization . Lubricants: Meaning , type and theory of lubricants, properties of a good lubricants, Flash and fire point and cloud point, emulsification number, viscosity. Paints and Varnishes : Meaning,
12.	FUELS, FIRE EXTINGUISHERS AND EXPLOSIVES	ingredients and characteristics of good paints and varnishes, their engineering applications calorific value, Determination of a solid fuel by bomb calorimeter , octane and octane number. Proximate analysis of fuel, its utility, crude petroleum, products of fractional distillation .Fire extinguishers – Description and use.
13.	POLLUTION AND CONTROL	Explosives – Meaning, types, characteristic and use of explosives. Name Dynamite, lead azide, T.N.T., Picric acid, R.D.X.Introduction and chemical toxicology, air and water pollution, control of air and water pollution. Harmful effect of different gases like carbon mono-oxide, carbon dioxide, sulphur dioxide, nitric oxide, nitrous and lead.



BHABHA UNIVERSITY, BHOPAL
DIPLOMA IN
MECHANICAL ENGINEERING

SEMESTER: **FIRST**
COURSE CODE: **1001**
NAME OF COURSE: **CHEMISTRY**

SCHEME: **2018**

Practical: 2 Hrs. per week

LIST OF EXPERIMENTS

S.NO.	NAME OF THE EXPERIMENT
1	To identify one Anion and Cation in a given sample.
2.	Determination of flash point and fire point of a given sample of oil by Abel's apparatus.
3.	Determination of viscosity by Red Wood Viscometer no. 1 and no.2.
4.	Redoximetry Titration : Percentage of Iron in given sample of alloy. Determination of strength of ferrous ammonium Determination of strength of anhydrous ferrous sulphate and ferrous sulphate.
5	Determination of hardness of water by : a. EDTA Method and Soap Solution Method
6	Determination of solid content in the given
7.	Determination of percentage of moisture in the given sample of coal by proximate analysis.



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SEMESTER: **FIRST**
COURSE CODE: **1001**
NAME OF COURSE: **CHEMISTRY**

SCHEME: **2018**

REFERENCES

1. PHYSICAL CHEMISTRY
– BAHL AND TULI
2. INORGANIC CHEMISTRY
– SATYAPRAKASH
3. MODERN TEXT BOOK OF APPLIED CHEMISTRY
– DR. G. C. SAXENA, JAIN PRAKASHAN, INDORE
4. APPLIED CHEMISTRY
- DR. G. C. SAXENA, DEEPAK PRAKASHAN, GWALIOR
5. APPLIED CHEMISTRY
– SHRIVASTAVA & SINGHAL, PBS PUBLICATION, BHOPAL
6. ENGINEERING CHEMISTRY
– UPPAL
7. ENGINEERING CHEMISTRY
– RAO AND AGARWAL
8. ENGINEERING CHEMISTRY
– P.C. JAIN
9. POLYMER CHEMISTRY
– O.P. MISHRA
10. APPLIED CHEMISTRY
– H.N. SAHNI, DEEPAK PRAKASH



BHABHA UNIVERSITY, BHOPAL
DIPLOMA IN
MECHANICAL ENGINEERING

SEMESTER: **FIRST**

COURSE CODE: **MA-1001**

NAME OF COURSE: **MATHEMATICS**

RATIONALE

Mathematics forms backbone for all technologies and hence occupies an important place in the curriculum of polytechnic education. The subject is equally important for the future self development of Polytechnic students. In designing the curriculum for foundation course the admission level to Polytechnics has been considered as 10th Board examination and mathematical needs of Technical subject have been given due consideration. To understand difficult concepts in higher engineering courses and to solve many problems of design and development a good background in mathematics is necessary. In view of this course is required for engineering diploma programmes.



BHABHA UNIVERSITY, BHOPAL
DIPLOMA IN
MECHANICAL ENGINEERING

SEMESTER: FIRST

COURSE CODE:MA-1001

NAME OF COURSE: MATHEMATICS

SCHEME:2018

Lectures: 6 Hrs. per week

SCHEME OF STUDIES

S. No.	Topic
1	ALGEBRA
2	TRIGONOMETRY
3	MATRIX
4	CO-ORDINATE GEOMETRY
5	STATISTICS
6	DIFFERENTIAL CALCULUS
7	INTEGRAL CALCULUS
8	VECTOR ALGEBRA



BHABHA UNIVERSITY, BHOPAL
DIPLOMA IN
MECHANICAL ENGINEERING

SEMESTER: **FIRST**

COURSE CODE: **MA-1001**

NAME OF COURSE: **MATHEMATICS**

SCHEME: 2018

SEMESTER: **FIRST**

Lectures: 6 Hrs. per week

COURSE CONTENTS

S.NO.	TOPIC	CONTENTS
1	ALGEBRA	1 Permutation Meaning of factorial n- Permutation of 'n' dissimilar thing taken 'r' at a time, 1 Combination - Combination of n dissimilar things taken 'r' at a time, 1 Binomial Theorem - Statement of the theorem for positive integer - General Term, Middle term, Constant term 1 Partial Fractions - Define a proper-improper fraction - Break a fraction into partial fraction whose denominator contains Linear, Repeated linear and Non repeated quadratic factors. 2 Determinant - Concept & principles of determinants - Properties of determinant - Simple examples. 2 Complex Numbers - Algebra of Complex Numbers - Polar form
2	TRIGONOMETRY	2 Allied angles. 2.2 Trigonometrical ratios of sum and difference of angles, (Only statement) Sum and difference of trigonometric ratios (C-D formula) 2.3

S.NO.	TOPIC	CONTENTS
3.	MATRIX	2.4 Multiple angles (Only double angle and half angle) 2.5 Properties of triangle (without proof) 3.1 Definition of Matrix. 3.2 Types of Matrix. - Row, Column, Square, Unit, Upper and lower triangular, Symmetric & Skew Symmetric, Singular and non Singular Matrices. 3.3 Adjoint of a Matrix. 3.4 Inverse of a Matrix.
4.	CO-ORDINATE GEOMETRY	4.1 Co-ordinate System : Cartesian and Polar. 4.2 Distance, Division, Area of a triangle. 4.3 Locus of a point and its equation. 4.4 Slope of St. Line - Angle between two St. lines. - Parallel and perpendicular St. lines. 4.5 Standard and general equation of St. line. Point of intersection of two st lines.
5	STATISTICS	5.1 Measures of Central tendency (Mean, Mode, Median) 5.2 Measures of Dispersion (Mean deviation, standard deviation)
6.	DIFFERENTIAL CALCULUS	6.1 Define constant, variable, function. 6.2 Value of the function 6.3 Concept of limit of a function. 6.4 Definition and concept of differential coefficient as a limit. 6.5 Standard results. 6.6 Derivatives of sum, difference, product, quotient of two functions. 6.7 Diff. coeff. of function of a function. 6.8 Diff. coeff. of implicit function. 6.9 Logarithmic Differentiation. 6.10 Differential coeff. of Parametric function.
7.	INTEGRAL CALCULUS	7.1 Definition as a inverse process of differentiation 7.2 Standard Results (including inverse function) 7.3 Methods of Integration - Substitution

S.NO.	TOPIC	CONTENTS	
8.	VECTOR ALGEBRA		- Integration by parts - Breaking up into partial fraction
		7,4	Concept of Definite Integral
		8.1	Concept of Vector and Scalar Quantities. 20
			Different types of vectors.
		8,2	Addition and subtraction of vectors.
		8,3	Components of a vector
			Multiplication of two vectors
		8.4	- Scalar Product
		8.5	- Vector Product
			- Applications (Work done, power & reactive power)



BHABHA UNIVERSITY, BHOPAL

DIPLOMA IN MECHANICAL ENGINEERING

SEMESTER: FIRST

SCHEME:2018

COURSE CODE: MA-1001

NAME OF COURSE: MATHEMATICS

REFERENCES

1. Mathematics for Polytechnics Vol. I and II
 - Prepared by T.T.T.I. Bhopal
2. Differential Calculus
 - Gorakh Prasad
3. Integral Calculus
 - Gorakh Prasad
4. Co-ordinate Geometry
 - S.L. Loni
5. Engineering Mathematics (M.P. Hindi Granth Akadami)
 - Dr. S.K. Chouksey & Manoj Singh
6. Mathematical Statistics
 - Ray and Sharma
7. Higher Engineering Mathematics
 - B.S. Grewal



BHABHA UNIVERSITY, BHOPAL

**DIPLOMA IN
MECHANICAL ENGINEERING**

SEMESTER: **FIRST**

SCHEME:2018

COURSE CODE: PA-1001

NAME OF COURSE: **PROFESSIONAL ACTIVITIES & HUMAN VALUE**

Practical: 2 Hrs. per week

RATIONALE

Professional Activities is not a descriptive course, as per conventional norms; therefore specific content for this course cannot be prescribed. It is a group of open-ended activities; where in variety of tasks are to be performed, to achieve objectives. However general guidelines for achieving the target and procedure for its assessment are given under the course content.

As the student has to practice this course in all the six semesters, the guidelines given therein are common and applicable to each semester.

OBJECTIVES:

- To allow for professional development of students as per the demand of engineering profession.
- To provide time for organization of student chapter activities of professional bodies) i.e. Institute of engineers, ISTE or Computer Society of India etc.)
- TO allow for development of abilities in students for leadership and public speaking through organization of student's seminar etc.
- To provide time for organization of guest lectures by expert engineers/eminent professionals of industry.
- To provide time for organization of technical quiz or group discussion or any other group activity.
 - To provide time for visiting library or using Internet.
 - To provide time for group discussion or solving case studies.
 - To provide time for personality development of students.
- To provide time for working for social cause like awareness for environmental and ecology etc.

DETAILED INSTRUCTIONS TO CONDUCT PROFESSIONAL ACTIVITIES:

- A. Study hours, if possible should be given greater time slot with a minimum of two hrs/week to a maximum of four hrs/week.
- B. This course should be evaluated on the basis of grades and mark sheet of students, should have a separate mention of the grade awarded. There will be no pass/fail in professional activities (PA).
- C. Following grade scale of evaluation of performance in PA has been established

Grade Level of performance

- A Excellent
- B Good
- C Fair

Fundamental of Electrical & Civil Engineering Practice

FU: 1001

PRACTICAL EXERCISES

I. Chain surveying

- i)
 - a) Ranging a line
 - b) Chaining a line and recording in the field book
 - c) Taking offsets - perpendicular and oblique (with a tape only)
 - d) Setting out right angle with a tape
- ii) Chaining of a line involving reciprocal ranging
- iii) Chaining a line involving obstacles to ranging
- iv) Chain Survey of a small area.

II. Compass Surveying

- a) Study of prismatic compass
- b) Setting the compass and taking observations
- c) Measuring angles between the lines meeting at a point

III. Levelling

- i)
 - a) Study of dumpy level and leveling staff
 - b) Temporary adjustments of various levels
 - c) Taking staff readings on different stations from the single setting and finding differences of level between them
- ii) To find out difference of level between two distant points by shifting the instrument
- iii) Longitudinal and cross sectioning of a road/railway/canal
- iv) Setting a gradient by dumpy and auto-level

IV. Plane Table Surveying

i)

- a) Study of the plane table survey equipment
- b) Setting the plane table
- c) Marking the North direction
- d) Plotting a few points by radiation method

ii)

- a) Orientation by - Trough compass - Back sighting
- b) Plotting few points by intersection, radiation and resection method

iii) Traversing an area with a plane table (at least five lines)

V. Layout of Buildings (from given drawing of two room residential building) by use of surveying instruments.

RECOMMENDED BOOKS

- 1. A Text Book of Surveying by Kochar, CL; Katson Publishing House, Ludhiana,
- 2. Surveying and Leveling by Kanetkar, TP and Kulkarni, SV; AVG Parkashan, Poona
- 3. Surveying –I by Mahajan, Sanjay; Tech. Publication, Delhi
- 4. Surveying and Leveling by Punmia, BC; Standard Publishers Distributors, Delhi.
- 5. E-books/e-tools/relevant software to be used as recommended by AICTE/UBTE/NITTTR, Chandigarh.

LIST OF PRACTICALS

- 1. Study of green energy
- 2. Operation and use of measuring instruments viz voltmeter, ammeter, CRO, Wattmeter, multi-meter .
- 3. Determination of voltage-current relationship in a dc circuit under specific physical conditions and to draw conclusions.
- 4. Measurement of resistance of an ammeter and a voltmeter

5. Verification of dc circuits:
 - a. Thevenin's theorem,
 - b. Norton's theorem,
6. Observation of change in resistance of a bulb in hot and cold conditions, using voltmeter and ammeter.
7. Verification of Kirchhoff's Current and Voltage Laws in a dc circuit
8. Charging and testing of a lead - acid storage battery.
9. Measurement of power and power factor in a single phase R-.L-.C. circuit and calculation of active and reactive powers in the circuit.
10. Verification of line to line and line to neutral voltages and current in star and delta connections.
11. Study of Wiring system of Public Buildings & Residential buildings.

Note: Visit to a nearby Power Station(s) may be arranged to demonstrate various aspects of subject.

RECOMMENDED BOOKS

1. Basic Electrical Engineering by PS Dhogal, Tata McGraw-Hill Education Pvt Ltd., New Delhi.
2. Experiments in Basic Electrical Engineering by SK Bhattacharya, KM Rastogi; New Age International (P) Ltd.; Publishers New Delhi.
3. Electrical Technology by BL Theraja, S Chand and Co, New Delhi.
4. Basic Electrical Engineering by JB Gupta; SK Kataria and Sons, New Delhi.
- 5.. Basic Electrical Engineering by Asfaque Husain, Jain Book Depot, New Delhi